1) a)

$$\frac{1}{\frac{\sqrt{2}}{2}-0}\int_0^{\frac{\sqrt{2}}{2}} \pi x\cos(\pi x^2)\,\mathrm{d}x = \sqrt{2}\left[\frac{1}{2}\sin(\pi x^2)\right]_0^{\frac{\sqrt{2}}{2}} = \sqrt{2}\left[\frac{1}{2}-0\right] = \frac{\sqrt{2}}{2}$$

1) b)

$$\frac{1}{\frac{\pi}{3}-\frac{\pi}{4}}\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \operatorname{cosec}^2 x\,\mathrm{d}x = \frac{12}{\pi}\left[-\cot x\right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} = \frac{12}{\pi}\left[-\frac{\sqrt{3}}{3}-(-1)\right] = \frac{12-4\sqrt{3}}{\pi}$$

1) c)

$$\frac{1}{1-0}\int_0^1 \frac{x^2}{5x^3+1}\,\mathrm{d}x = \left[\frac{1}{15}\ln|5x^3+1|\right]_0^1 = \left[\frac{1}{15}\ln 6-0\right] = \frac{1}{15}\ln 6$$

1) d)

$$\frac{1}{\frac{\pi}{4}-0}\int_0^{\frac{\pi}{4}} \sin^2 4x\,\mathrm{d}x$$

Since $\cos 2A = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos 2A$ so that

$$\frac{4}{\pi}\int_0^{\frac{\pi}{4}} \sin^2 4x\,\mathrm{d}x = \frac{4}{\pi}\int_0^{\frac{\pi}{4}} \left(\frac{1}{2}-\frac{1}{2}\sin 8x\right)\mathrm{d}x = \frac{4}{\pi}\left[\frac{1}{2}x+\frac{1}{16}\cos 8x\right]_0^{\frac{\pi}{4}} = \frac{4}{\pi}\left[\frac{1}{16}+\frac{\pi}{8}-\frac{1}{16}\right] = \frac{1}{2}$$

1) e)

$$\frac{1}{5-2}\int_2^5 2^x\,\mathrm{d}x = \frac{1}{3}\left[\frac{1}{\ln 2}2^x\right]_2^5 = \frac{1}{3}\left[\frac{1}{\ln 2}2^5-\frac{1}{\ln 2}2^2\right] = \frac{28}{3\ln 2}$$

1) f)

$$\frac{1}{\frac{\pi}{12}-0}\int_0^{\frac{\pi}{12}} \sin 2x\cos^3 2x\,\mathrm{d}x = \frac{12}{\pi}\left[-\frac{1}{8}\cos^4 2x\right]_0^{\frac{\pi}{12}} = \frac{12}{\pi}\left[-\frac{9}{128}--\frac{1}{8}\right] = \frac{21}{32\pi}$$

2)

Use Integration by Parts:

$$u = x \qquad \frac{dv}{dx} = e^{-3x}$$

$$\frac{du}{dx} = 1 \qquad v = -\frac{1}{3}e^{-3x}$$

$$\therefore \int xe^{-3x}dx = -\tfrac{1}{3}xe^{-3x} + \tfrac{1}{3}\int e^{-3x}\,dx = -\tfrac{1}{3}xe^{-3x} - \tfrac{1}{9}e^{-3x}(+c)$$

So that:

$$\frac{1}{a-0}\int_0^a xe^{-3x}\,dx = \frac{1}{a}\left[-\frac{1}{3}xe^{-3x} - \frac{1}{9}e^{-3x}\right]_0^a = \frac{1}{a}\left[\left(-\frac{1}{3}ae^{-3a} - \frac{1}{9}e^{-3a}\right) - \left(0 - \frac{1}{9}\right)\right]$$

$$= -\frac{1}{3}e^{-3a} - \frac{1}{9a}e^{-3a} + \frac{1}{9a}$$

3)

$$\frac{1}{8-0}\int_0^8 \left(\frac{2}{\sqrt{3x+1}} + k\right)dx = \frac{1}{8}\int_0^8 \left(2(3x+1)^{-\frac{1}{2}} + k\right)dx = \frac{1}{8}\left[\frac{4}{3}\sqrt{3x+1} + kx\right]_0^8$$

$$= \tfrac{1}{8}\left[\left(\tfrac{4}{3}\times 5 + 8k\right) - \left(\tfrac{4}{3} + 0\right)\right]$$

$$= \frac{5}{6} + k - \frac{1}{6}$$

$$\Rightarrow \tfrac{5}{6} + k - \tfrac{1}{6} = k + \tfrac{2}{3} = 2$$

$$\therefore k = \frac{4}{3}$$

4)

$$\frac{1}{3-0}\int_0^3 \sqrt{x+1}\,dx = \frac{1}{3}\int_0^3 (x+1)^{\frac{1}{2}}\,dx = \frac{1}{3}\left[\frac{2}{3}(x+1)^{\frac{3}{2}}\right]_0^3 = \frac{1}{3}\left[\frac{16}{3} - \frac{2}{3}\right] = \frac{14}{9}$$

$$\frac{1}{k-0}\int_0^k \frac{3}{7}x^2\,dx = \frac{1}{k}\left[\frac{1}{7}x^3\right]_0^k = \frac{1}{k}\left[\frac{1}{7}k^3\right] = \frac{1}{7}k^2$$

And so:

$$\frac{14}{9} = \frac{1}{2}\left(\frac{1}{7}k^2\right)$$

$\frac{196}{9} = k^2$ so $k = \pm\frac{14}{3}$

$$\therefore k = \frac{14}{3}$$

5) a)

$\frac{30t}{t+5} = \frac{30(t+5)-150}{t+5} = 30 - \frac{150}{t+5}$

$$\frac{1}{20-0}\int_0^{20} \frac{30t}{t+5}\,\mathrm{d}t = \frac{1}{20}\int_0^{20}\left(30 - \frac{150}{t+5}\right)\mathrm{d}t = \frac{1}{20}\left[30t - 150\ln|t+5|\right]_0^{20}$$

$= \frac{1}{20}[600 - 150\ln 25 - (0 - 150\ln 5)] = 17.9$ ms^{-1} (as req).

5) b)

$a = \frac{30(t+5)-30t}{(t+5)^2} = \frac{150}{(t+5)^2}$

$$\frac{150}{20-0}\int_0^{20}(t+5)^{-2}\,\mathrm{d}t = \frac{15}{2}\left[-(t+5)^{-1}\right]_0^{20}$$

$= \frac{15}{2}\left[-\frac{1}{25} - \left(-\frac{1}{5}\right)\right] = \frac{6}{5} = 1.2$ ms^{-2}

6) a)

Use Integration by Parts:

$$u = \ln x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \qquad v = x$$

$$\therefore \int \ln x\,\mathrm{d}x = x\ln x - \int x \times \frac{1}{x}\mathrm{d}x = x\ln x - x(+c)$$

So that:

$$\frac{1}{4-1}\int_1^4 \ln x\,\mathrm{d}x = \frac{1}{3}\left[x\ln x - x\right]_1^4 = \frac{1}{3}[(4\ln 4 - 4) - (0-1)] = -1 + \frac{4}{3}\ln 4$$

6) b)

$-1 + \frac{4}{3}\ln 4 + c = \frac{4}{3}\ln 4 \therefore c = 1$

7)

$$\frac{1}{\frac{\pi}{2}-0}\int_0^{\frac{\pi}{2}} \sin x\,\mathrm{d}x = \frac{2}{\pi}\left[-\cos x\right]_0^{\frac{\pi}{2}} = \frac{2}{\pi}\left[-\cos\frac{\pi}{2} - (-\cos 0)\right] = \frac{2}{\pi}(0+1) = \frac{2}{\pi}$$

8)

$$\frac{1}{2\pi-0}\int_0^{2\pi}(4\cos x)^2\,\mathrm{d}x=\frac{16}{2\pi}\int_0^{2\pi}\cos^2 x\,\mathrm{d}x$$

Since $\cos 2A=2\cos^2 A-1$ then $\cos^2 A=\frac{1}{2}+\frac{1}{2}\cos 2A$ so that

$$\frac{16}{2\pi}\int_0^{2\pi}\cos^2 x\,\mathrm{d}x=\frac{8}{\pi}\int_0^{2\pi}\left(\frac{1}{2}+\frac{1}{2}\sin 2x\right)\mathrm{d}x=\frac{8}{\pi}\left[\frac{1}{2}x-\frac{1}{4}\cos 2x\right]_0^{2\pi}$$

$$=\frac{8}{\pi}\left[\left(\pi-\tfrac{1}{4}\right)-\left(0-\tfrac{1}{4}\right)\right]=8$$

$$\therefore\sqrt{\tfrac{1}{2\pi-0}\textstyle\int_0^{2\pi}(4\cos x)^2\,\mathrm{d}x}=2\sqrt{2}$$

9) a)

Since $\cos 2A=2\cos^2 A-1$ then $\cos^2 A=\frac{1}{2}+\frac{1}{2}\cos 2A$ so that

$$\cos^4\theta=\left(\tfrac{1}{2}+\tfrac{1}{2}\cos 2\theta\right)\left(\tfrac{1}{2}+\tfrac{1}{2}\cos 2\theta\right)=\tfrac{1}{4}(1+2\cos 2\theta+\cos^2 2\theta)$$

$$=\tfrac{1}{4}\left(1+2\cos 2\theta+\tfrac{1}{2}+\tfrac{1}{2}\cos 4\theta\right)$$

$$=\tfrac{1}{4}\left(\tfrac{3}{2}+2\cos 2\theta+\tfrac{1}{2}\cos 4\theta\right)$$

$$=\tfrac{1}{8}(3+4\cos 2\theta+\cos 4\theta)$$ as req.

9) b)

$$v^2=a^2\cos^4(kt)$$

$$\frac{1}{\frac{\pi}{k}-0}\int_0^{\frac{\pi}{k}}a^2\cos^4(kt)\,\mathrm{d}x=\frac{a^2k}{\pi}\int_0^{\frac{\pi}{k}}\left(\frac{1}{8}(3+4\cos 2kt+\cos 4kt)\right)\mathrm{d}x$$

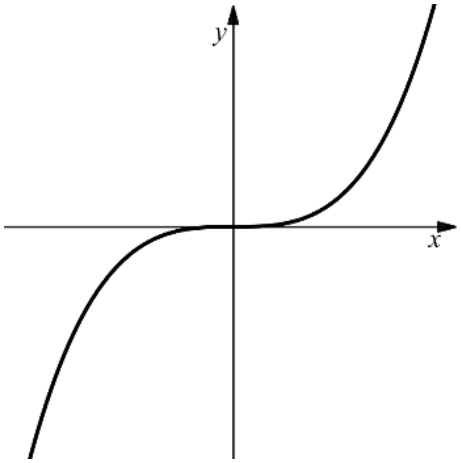
$$=\frac{a^2k}{8\pi}\left[3t+\frac{4}{2k}\sin 2kt+\frac{1}{4k}\sin 4kt\right]_0^{\frac{\pi}{k}}$$

$$=\tfrac{a^2k}{8\pi}\left[\left(\tfrac{3\pi}{k}+0+0\right)-(0+0+0)\right]=\tfrac{3a^2}{8}$$

$$\therefore\sqrt{\tfrac{1}{\frac{\pi}{k}-0}\textstyle\int_0^{\frac{\pi}{k}}a^2\cos^4(kt)\,\mathrm{d}x}=\sqrt{\tfrac{3a^2}{8}}=\tfrac{\sqrt{3}a}{\sqrt{8}}=\tfrac{\sqrt{3}a}{2\sqrt{2}}\ \text{ms}^{-1}$$ as req.

10) a)

Since $\sinh x$ is an odd function, then since n is odd, $\sinh^n x = (\sinh x)^n$ is also an odd function.



$\therefore$ the mean value is 0

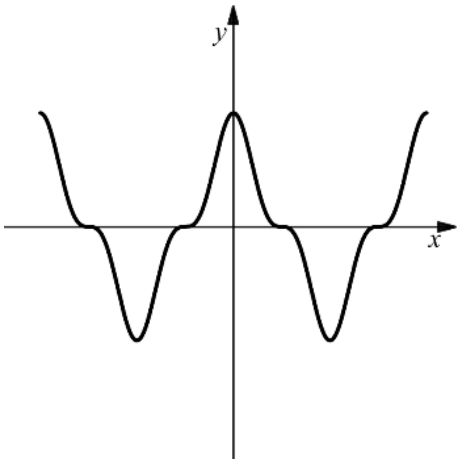
10) b)

$\cosh(\mathrm{i}x) = \frac{\mathrm{e}^{\mathrm{i}x}+\mathrm{e}^{-\mathrm{i}x}}{2} = \frac{\cos x+\mathrm{i}\sin x+\cos(-x)+\mathrm{i}\sin(-x)}{2}$

$= \frac{\cos x+\mathrm{i}\sin x+\cos x-\mathrm{i}\sin x}{2}$

$= \cos x$

Since $\cos x$ is an even function, then since n is odd, $\cos^n x = (\cos x)^n$ is also an even function. So its graph is symmetrical about $x = 0$



So the graph of $y = (\cos x)^n$ for $-k\pi \leq x \leq k\pi$ for positive integer k has the same shape/area above and below $y = 0$

$\therefore$ the mean value is 0

10) c)

$\sinh(\mathrm{i}x) = \frac{\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x}}{2} = \frac{\cos x + \mathrm{i}\sin x - \cos(-x) - \mathrm{i}\sin(-x)}{2}$

$= \frac{\cos x + \mathrm{i}\sin x - \cos x + \mathrm{i}\sin x}{2}$

$= \mathrm{i}\sin x$

Since $\sin x$ is an odd function, then since n is odd, $\sin^n x = (\sin x)^n$ is also an odd function. Also since i^n is a constant, $(\mathrm{i}\sin x)^n$ is also an odd function.

$\therefore$ the mean value is 0

11) a)

$$\frac{1}{3-0}\int_0^3 \frac{1}{9+x^2}\,\mathrm{d}x = \frac{1}{3}\left[\frac{1}{3}\arctan\left(\frac{x}{3}\right)\right]_0^3 = \frac{1}{3}\left[\frac{\pi}{12} - 0\right] = \frac{\pi}{36}$$

11) b)

$$\frac{1}{\frac{1}{4}-0}\int_0^{\frac{1}{4}} \frac{1}{\sqrt{1-4x^2}}\,\mathrm{d}x = 4\left[\frac{1}{2}\arcsin(2x)\right]_0^{\frac{1}{4}} = 4\left[\frac{\pi}{12} - 0\right] = \frac{\pi}{3}$$

11) c)

$$\frac{1}{a-0}\int_0^a \frac{1}{x^2+7}\,\mathrm{d}x = \frac{1}{a}\left[\frac{1}{\sqrt{7}}\arctan\left(\frac{x}{\sqrt{7}}\right)\right]_0^a = \frac{1}{a}\left[\frac{1}{\sqrt{7}}\arctan\left(\frac{a}{\sqrt{7}}\right) - 0\right] = \frac{1}{a\sqrt{7}}\arctan\left(\frac{a}{\sqrt{7}}\right)$$

12)

$$\frac{1}{6-0}\int_0^6 \frac{1}{\sqrt{x^2-6x+16}}\,\mathrm{d}x = \frac{1}{6}\int_0^6 \frac{1}{\sqrt{7+(x-3)^2}}\,\mathrm{d}x$$

$$= \frac{1}{6}\left[\ln\left(x-3+\sqrt{7+(x-3)^2}\right)\right]_0^6$$

$$= \frac{1}{6}[\ln 7 - 0] = \frac{1}{6}\ln 7$$

13)

$$\frac{1}{a-3}\int_3^a \frac{1}{\sqrt{x^2-9}}\,\mathrm{d}x = \frac{1}{a-3}\left[\ln\left(x+\sqrt{x^2-9}\right)\right]_3^a = \frac{1}{a-3}\left[\ln\left(a+\sqrt{a^2-9}\right) - \ln 3\right]$$

$$= \frac{1}{a-3}\ln\left(\frac{a+\sqrt{a^2-9}}{3}\right)$$

14)

$$\frac{1}{4-0}\int_0^4 (12+8\,\text{sech}^2(0.5t))\,\text{d}t = \frac{1}{4}[12t+16\tanh(0.5t)]_0^4 = \frac{1}{4}[48+16\tanh(2)-0]$$

$= 15.9$ watts per square metre.

15)

$$\frac{1}{2-1}\int_1^2 (50+20\,\text{cosech}^2(0.2t))\,\text{d}t = [50t-100\coth(0.2t)]_1^2$$

$= 100-100\coth(0.4)-(50-100\coth(0.2))$

$= 293$ Lumens

16) a)

$$\frac{1}{12-0}\int_0^{12}\frac{18}{\sqrt{t^2+9}}\text{d}t = \frac{18}{12}\left[\ln\left(t+\sqrt{t^2+9}\right)\right]_0^{12} = \frac{3}{2}\left[\ln\left(12+\sqrt{144+9}\right)-\ln 3\right]$$

$= 3.14$ ms^{-1} (as req.)

16) b)

$$a = \frac{\text{d}v}{\text{d}t} = -18t(t^2+9)^{-\frac{3}{2}}$$

$$\frac{18}{12-0}\int_0^{12} -t(t^2+9)^{-\frac{3}{2}}\,\text{d}t = \frac{3}{2}\left[(t^2+9)^{-\frac{1}{2}}\right]_0^{12}$$

$= \frac{3}{2}\left[\frac{1}{3\sqrt{17}}-\frac{1}{3}\right] = -0.379$ ms^{-2}

∴ a mean deceleration of 0.379 ms^{-2}

17)

$$\frac{1}{a-0}\int_0^a \frac{1}{\sqrt{25-x^2}}\mathrm{d}x = \frac{1}{a}\left[\arcsin\left(\frac{x}{5}\right)\right]_0^a = \frac{1}{a}\left[\arcsin\left(\frac{a}{5}\right)-0\right] = \frac{1}{a}\arcsin\left(\frac{a}{5}\right)$$

Since $\cos\theta = \frac{a}{5}$ we have:

$$\sin\left(\frac{\pi}{2}-\theta\right) = \frac{a}{5}$$

Note that $\theta \in [0,\pi]$ so $\frac{\pi}{2}-\theta \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ which lies in the principal value of arcsin, so it is the principal value.

Hence

$$\cos\theta = \frac{a}{5} \Rightarrow \sin\left(\frac{\pi}{2}-\theta\right) = \frac{a}{5} \Rightarrow \frac{\pi}{2}-\theta = \arcsin\left(\frac{a}{5}\right)$$

So that:

$$\frac{1}{a}\arcsin\left(\frac{a}{5}\right) = \frac{1}{a}\left(\frac{\pi}{2}-\theta\right) = \frac{\pi}{2a}-\frac{1}{a}\theta$$

18)

$$\frac{x+2}{(x^2+2)(x-1)} \equiv \frac{A}{x-1}+\frac{Bx+C}{x^2+2}$$

$$x+2 = A(x^2+2)+(Bx+C)(x-1)$$

Let $x=1$: $\quad 3 = 3A \therefore A = 1$

Let $x=0$: $\quad 2 = 2A - C \therefore C = 0$

Let $x=2$: $\quad 4 = 6A + 2B \therefore B = -1$

$$\frac{x+2}{(x^2+2)(x-1)} \equiv \frac{1}{x-1}-\frac{x}{x^2+2}$$

$$\frac{1}{\frac{5}{2}-\frac{3}{2}}\int_{\frac{3}{2}}^{\frac{5}{2}} \frac{x+2}{(x^2+2)(x-1)}\mathrm{d}x = \int_{\frac{3}{2}}^{\frac{5}{2}}\left(\frac{1}{x-1}-\frac{x}{x^2+2}\right)\mathrm{d}x$$

$$= \left[\ln|x-1|-\frac{1}{2}\ln|x^2+2|\right]_{\frac{3}{2}}^{\frac{5}{2}} = \left(\ln\frac{3}{2}-\frac{1}{2}\ln\frac{33}{4}\right)-\left(\ln\frac{1}{2}-\frac{1}{2}\ln\frac{17}{4}\right)$$

$$= \ln\frac{3}{2}-\ln\frac{\sqrt{33}}{2}-\ln\frac{1}{2}+\ln\frac{\sqrt{17}}{2}$$

$$= \ln\frac{\sqrt{561}}{11}$$

$$\therefore k = \ln\frac{\sqrt{561}}{11}$$